IY4113 Milestone 1

Assessment Details	Please Complete All Details
Group	A
Module Title	IY4113 Applied Software Engineering using Object-Orientated Programming
Assessment Type	Assigment
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- [x] *I confirm that this assignment is my own work. Where I have referred to academic sources, I have provided in-text citations and included the sources in the final reference list.*

- [x] *Where I have used AI, I have cited and referenced appropriately.

Purpose of the Program

The program is a console-based application that helps a user to record, manage and analyse their transport journeys for one day. The program applies discounts to certain passenger type, calculates the prices with a set of pre made data sets and also has a daily spending caps which shows the user how much they have spent and the travel breakdown by zone, time of day and passenger category.

Core functionality:

· Input validation

- makes sure that all input values are correct

· Adding a journey

- User enters journey details including date, zones, passenger type and time band

· List journeys

- display all recorded journeys in full

· Filter journey

- allow user to filter journey by date, zone, passenger type, time band

· Price calculation

- calculates price by using number of zones crossed, discount price, price from the dataset

· Journey storage

- stores all journeys in memory using a collection whilst its running

· Menu option

- allows user to be able to choose the option they want to use until the exit the program

· Remove journey

- allows user to remove journey by using a unique identifier

· Resetting

- allowing the user to reset after confirmation

· Daily summary

outputs the total number of journeys, total cost, Average cost per journey, Most expensive journey

· Totals by Passenger type

- Shows spending and journey count per passenger category

Constraints

- Temporary Memory Storage – all data is lost after the user exits the program

- Zones must be in the stated range which is zone 1-5

- Restricted time bands- off peak and peak
- Only one user is allowed to run the program
- They it supports only one day/session only
- Its console based there is no user interface
- The passenger types are restricted to adult, student, child, senior citizen
- Price calculation should meet the provided dataset

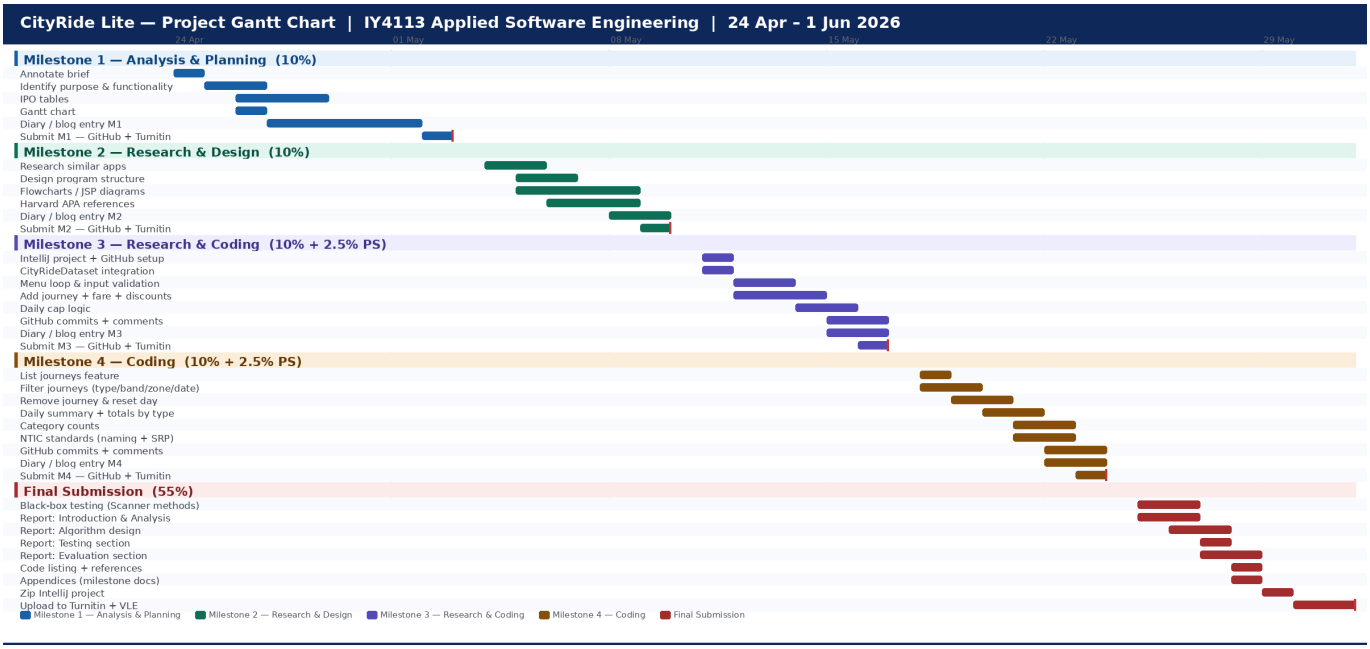
Input Process Output Table

Feature / Task	Inputs	Processing (what the system does)	Outputs
Start program	None program launch	Initialise data structures ArrayList, set counters and totals to zero	Main menu displayed
Display main menu	User menu choice 1–8	Validate input is numeric and within range; route to selected option	Selected feature executed or error message
Add journey	Date, from zone, to zone, passenger type, time band	Validate all inputs non-empty, correct format, valid ranges	Validated journey details OR error message
Calculate zones crossed	From zone, to zone	Apply formula: abs(toZone – fromZone) + 1	Number of zones crossed
Determine base price	Zones crossed, time band, dataset	Lookup correct price from provided dataset	Base price value
Apply discount	Base price , passenger type	Apply percentage discount based on passenger type	Discounted price
Apply daily cap	Running total per passenger type, discounted fare	Check if total exceeds cap; adjust price if needed; set future prices to £0 if cap reached	Final charged price
Store journey	All calculated journey data	Assign unique ID store in Array List	Journey saved successfully
List journeys	None menu selection	Retrieve all stored journeys format output	Display list of journeys with full details
Filter journeys	Filter criteria type, zone, date, time band	Search through stored journeys using criteria	Display filtered results
Remove journey	Journey ID	Validate ID exists remove journey recalculate totals	Confirmation or error message
Reset system	User confirmation yes/no	Clear all stored journeys and reset totals	System reset confirmation
Calculate daily summary	All stored journeys	Count journeys sum costs calculate average find highest fare	Total journeys, total cost, average cost, most expensive journey
Totals by passenger type	Stored journeys grouped by type	Calculate totals, count, and cap status per type	Summary per passenger type
Handle invalid input	Any incorrect input	Display error message and prompt user again	Corrected input entered
Exit program	Menu choice	Terminate program loop	Goodbye message

link to the original IPO table:
https://github.com/eugnx/T0466145_Eugene_4113/blob/master/IPO%20table%20.png

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Gantt Chart



link to the original Gantt chart:
https://github.com/eugnx/T0466145_Eugene_4113/blob/master/CityRide_Gantt.png

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Diary Entries

27/04/2026 -Diary Entry 1 – Understanding the Problem & Planning the Gantt Chart

Today I read through the assessment to understand the requirements and objectives. I broke the project into smaller tasks like planning, development, testing, and documentation. I then created a Gantt chart to organise my timeline and task order, considering dependencies between stages. This helped me visualise the project and plan my time effectively.Overall I now have a clear understanding of the task and a structured plan to follow.

28/04/2026 -Diary Entry 2 – Creating IPO Tables

Today I focused on analysing the inputs, processes, and outputs of the system by creating IPO tables. This helped me clearly understand how data flows through the program.

I identified that user inputs such as zones and passenger type must go through validation before any calculations are performed. I also mapped out how the system calculates zones crossed, determines base fares using the dataset, and applies discounts and caps.

One challenge I faced was ensuring that I included all major system functions in the IPO table without making it too complex. To solve this, I grouped related processes together and ensured each feature such as adding journeys or generating summaries was clearly represented.

Creating the IPO tables has made the program logic much clearer and will help guide my design in the next stage.

29/04/2026 -Diary Entry 2 – Planning and Gantt Chart,

Today, I focused on planning the project's development by creating a Gantt chart, which helped me organise the necessary tasks within the given timeline.

I broke the project down into stages: analysis, design, coding, testing, and final submission. Allocating time to each stage will help me stay on track and avoid last-minute pressure.

One risk I considered was spending too long on coding and neglecting documentation, which carries a significant portion of the marks. To address this, I ensured my plan balanced both areas appropriately.

Overall, this planning process has provided me with a clear roadmap for completing the project and will support effective time management.
